

MAGNETOM

System

Replacement Instructions

Magnet System

Aera

Skyra

Avanto fit

Skyra fit

Sola Fit

Vida Fit

Avanto Fit

Skyra Fit

M7-000.841.17.18.02 | 02.22

4 MSUP

4.1 Replacing the Magnet Supervisory Unit (MSUP 2306)

4.1.1 Requirements

This procedure is applicable to **OR98** and **OR99/OR103** systems.

i

The complete MSUP 2306 should only be exchange when instructed to do so by the HSC.

4.1.1.1 Safety information

i

This procedure can be performed with the magnet at field.

 **CAUTION**

The replacement procedure disables the magnet emergency shutdown system at the MSUP 2306.

Failure to adhere to safety precautions can result in serious bodily injury or property damage.

Connect the ERDU Test Load.

The magnet can only be quenched using the Magnet Stop button on the ERDU Test Load.

Exclude all non-service personnel and all magnetic materials / tools from the magnet area.

Close the door of the RF room.

4.1.1.2 Estimated completion time and number of persons

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	30 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

Tab. 44: Completion time

4.1.1.3 Tools

Non-magnetic tool kit.

ERDU test load kit. (part no **03861197**)

Laptop for data confirmation.

4.1.1.4 Spare parts

i

The 2306B MSUP (**10892850**) is an upgraded replacement for the 2306 MSUP (**10523587**).

MSUP 2306B tested (part no **10892850**)

Tamper-proof seal. (part no **8100831**)

4.1.2 Preparatory Work Steps

4.1.2.1 General

The **MSUP 2306** is a **Field Replaceable Unit**. It contains 2 assemblies:

The **MSUP 2306 controller assembly** is attached to the underside of the lid and is a **replaceable** spare part.  Fig. 527 (Pos. 1)

The **ERDU Test Load assembly** is mounted in the base of the MSUP and is **not separately replaceable**. If faulty, the **complete MSUP 2306 unit** must be replaced.  Fig. 527 (Pos. 3)

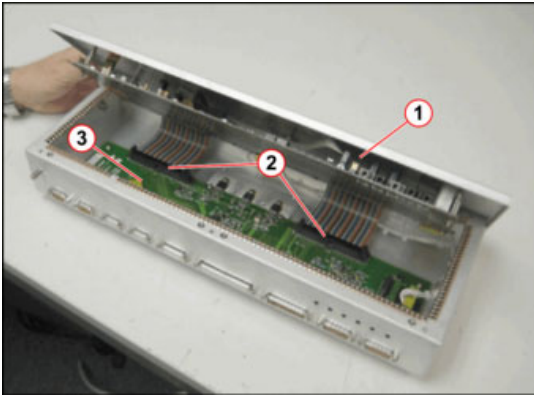


Fig. 527: MSUP 2306 complete unit

- 1 MSUP controller assembly (replaceable part)
- 2 Ribbon cables
- 3 ERDU assembly (not replaceable)

Remove the appropriate magnet cover panels.

4.1.2.2 MSUP 2306 positioning

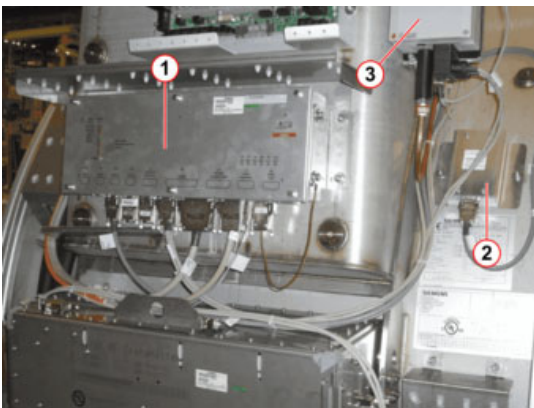


Fig. 528: MSUP 2306 position (static systems)

- 1 MSUP
- 2 ERDU battery
- 3 Pressure gauge

The MSUP 2306 is located behind the magnet service access covers. The unit and battery box are mounted on the magnet OVC. The battery is located to the right of the MSUP 2306.

4.1.3 Data confirmation after replacement

Ensure the i-Button and magnet data are captured from the original MSUP.

This data is then checked if it reads correctly on the new **MSUP**.

4.1.3.1 Capturing the iButton menu on the MSUP using a laptop

- 01 Turn on the laptop and open the **Terminal emulation software**, (eg HyperTerminal).

Set bits per second to 9600.

Set data bits to 8.

Set parity to 1.

Set flow control to none.

Set Emulation to VT100 (File:Properties:Settings)

- 02 **Connect** the laptop to connector **X6** on the MSUP **via RS232 cable**.

- 03 Press **ESC**.

- 04 Type **L0 6932** <enter>.

- 05 Type **R** <enter>.

- 06 Press **ESC** to display the **MSUP Command input screen**.

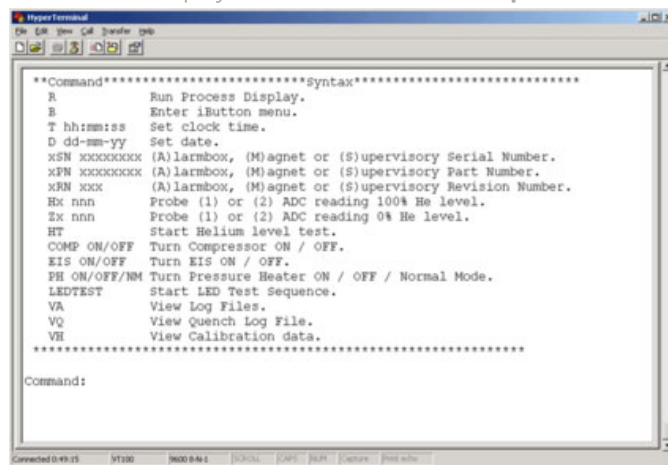


Fig. 529: MSUP Command Input screen (example)

- 07 Type **B** <enter> this displays the **iButton menu**.

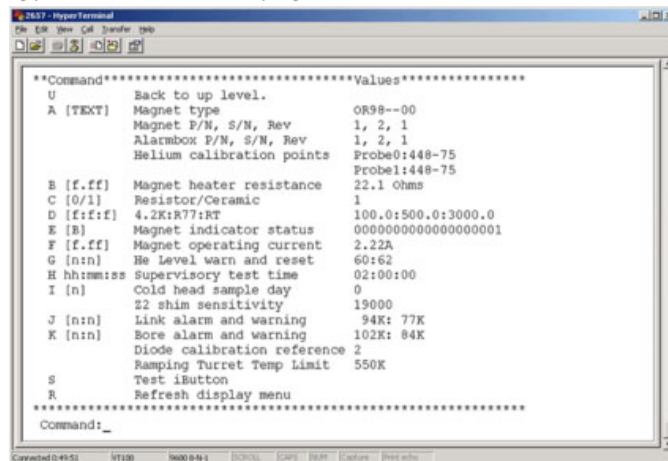


Fig. 530: MSUP 2306 iButton menu (example)

- 08 **Capture the screenshot.** ➤ Capturing a screenshot

10 Type **R** <enter>.

4.1.3.2 Capturing the data on the host

If you are unable to connect a laptop to the MSUP then most of the data can be obtained at the Host.

On the Host navigate to the Magnet & Cooling/Magnet Setup

Note down all the values displayed. The easiest way to do this is via taking a screen shot.



Fig. 531: Example of Numaris X SeSo/Magnet & Cooling/Magnet setup

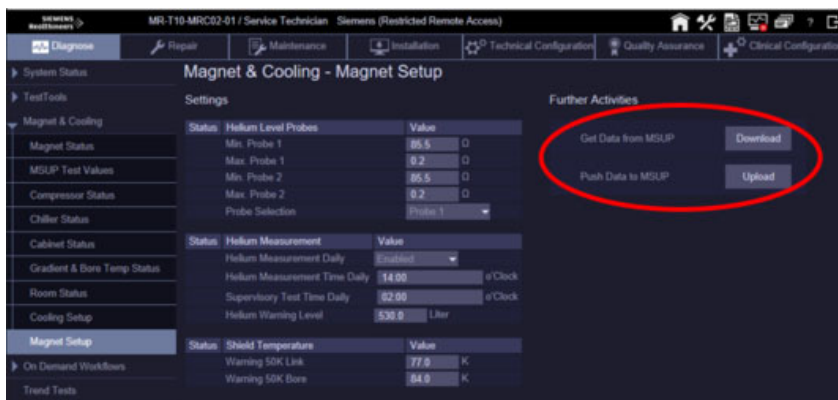


Fig. 532: Uploading data to the MSUP on Systems with Numaris X

4.1.3.3 Capturing a screenshot

Capture a **screenshot** of the **MSUP terminal emulation screen**:

- Press **Ctrl: Alt:Prt Sc.**
- Go to: **Start:Programs:Accessories:**and open Paint.
- Press **Ctrl:V** to paste screenshot.
- Name** and save as a **.jpg** or **.bmp**.

Name the file DD (**Day**), MM (**Month**), YYYY (**Year**), TTTT (**Time:24 hour format**).

i.e. iButtonDDMMYYYYTTTT: MSUPDDMMYYYYTTTT.

4.1.4 Connecting the ERDU Test Load (ETL)

WARNING

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorised personnel must be excluded from the imaging suite.

Secure the room and prevent unauthorised access.

WARNING

It is not possible to quench the magnet using the normal system MAGNET STOP buttons while the ERDU Test Load is connected to the MSUP and magnet.

The magnet may be quenched only by using the MAGNET STOP button on the ERDU Test Load.

Restrict access to the magnet accordingly.

CAUTION

The ERDU Test Load is slightly ferromagnetic.

The ETL is attracted by the magnetic force and will be projected into the magnet bore, resulting in injury or damage.

Keep the ETL as far away as possible from the magnet bore.

Check that the **ERDU Test Load Kit** is complete with all the listed parts present:

i

For this procedure, items 1 - 6, and 11 are required. Item 7 may be required depending on outlet socket.

When connected, the ETL acts as an ERDU button.

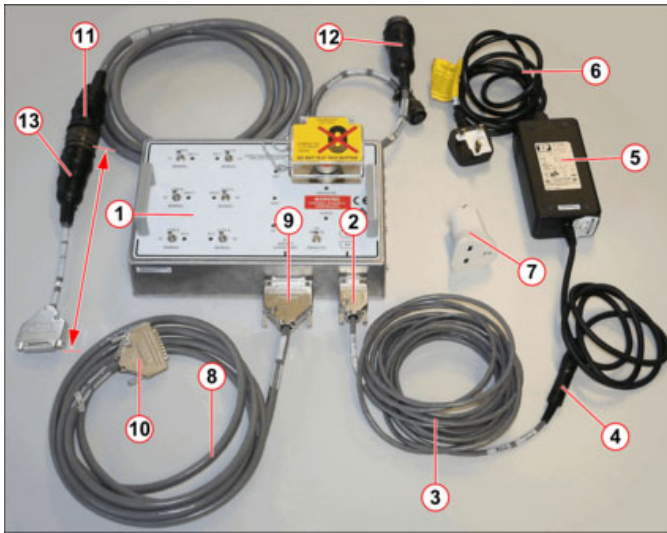


Fig. 533: ERDU Test load kit (2661)

- 1 ERDU Test load 3 (ETL)
- 2 PSU output extension lead to ETL X3 or X4
- 3 PSU output extension lead
- 4 PSU output extension lead connector to PSU
- 5 Power Supply Unit (PSU)
- 6 AC supply lead
- 7 International adapter - AC supply
- 8 MSUP - ETL cable
- 9 25-pin connector to ETL X1
- 10 25-pin connector to MSUP X1
- 11 41-pin connector, magnet turret
- 12 Adapter for OR64 / OR70, 41-pin connector to 10-pin magnet turret connector 'B' (optional ETL3 2661)
- 13 Adaptor cable B (optional for OR105, OR122, OR97)

Connect the **ETL PSU** to a convenient AC outlet **outside of the magnet room**, and switch on.  Fig. 533 (Pos. 5)

i

Ensure that the PSU ETL switch mode is as far away from the magnet as possible (magnetic field < 50 Gauss).

i

The ETL will sound a continuous buzzing alarm, until the ETL is disconnected.

Connect the ETL switch mode PSU to the ETL **X3** or **X4** using the extension cable.  Fig. 533 (Pos. 3)
Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.

WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

Ensure that the MAGNET STOP button on the ETL is NOT OPERATED!
DO NOT operate the MAGNET STOP button on the ERDU Test Load unless an emergency magnet quench is required.

Set **SH** for correct magnet type. For this magnet, set the load to (**OR105/122**).  Fig. 534



Fig. 534: ERDU Test Load (front panel)

Set all **HIGH R / LOW R** switches on the ETL to **NORMAL**.

WARNING

The magnet cannot be quenched!!

If the circuit between the 41 way connector on the magnet to the MSUP is disconnected, THERE IS NO ERDU FACILITY.

Reconnect X1 (or the 41-pin turret connector), WITHOUT DELAY.

As soon as the ETL is connected to the magnet, the test ERDU function on the ETL is operational.



Fig. 535: Turret - electrical connector

1 41 pin connector

Disconnect the existing lead from the 41-pin connector on the magnet turret.

IMMEDIATELY connect the 41-pin connector from the ETL to the magnet turret connection.  Fig. 533
(Pos. 11)

i

Keep the ETL as far away as possible from the magnet bore.

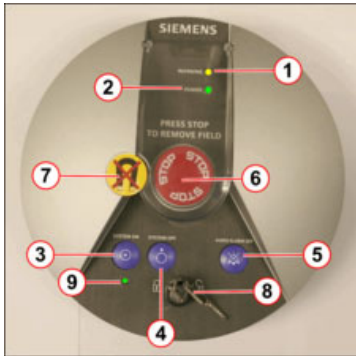


Fig. 536: Alarm box - LEDs and buttons

- 1 LED WARNING = yellow
- 2 LED POWER = green
- 3 Button - SYSTEM ON
- 4 Button - SYSTEM OFF
- 5 Button - AUDIO ALARM OFF
- 6 Button - MAGNET STOP (ERDU)
- 7 Tamper-proof seal
- 8 Power switch locking key
- 9 SYSTEM ON LED = green

Check that the alarm sounds and that the **WARNING** supply indicator lights up on the alarm box. Press **audio alarm OFF** to cancel the audible alarm.

Check that the **QH1**, **QH2**, **SH** and **SWITCH OK** and **SUPPLY** indicators **glow** on the ETL.

4.1.5 Removal

Shut down the system (**only necessary if the MSUP has revision level 2**).

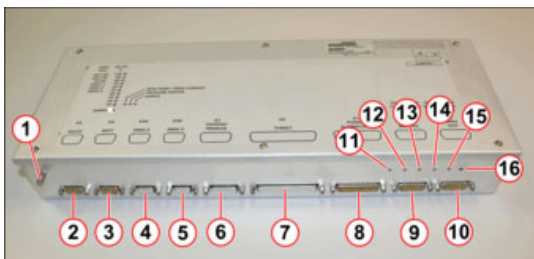


Fig. 537: MSUP 2306 indicators and input/output connections

- 1 Earth stud
- 2 X6 RS232
- 3 X3 Battery
- 4 X4A ERDU 2
- 5 X4B ERDU 3

- 6 X2 CH/DIAG/PRESS/ID
- 7 X5 TURRET
- 8 X1 POWER/ALARM/COMP
- 9 X7 ERDU NORMAL
- 10 X8 ERDU TEST (not connected when system is in use)
- 11 F/O TX1 U5
- 12 F/O RX1 U7
- 13 F/O TX2 U6
- 14 F/O RX2 U8
- 15 F/O CLK U9
- 16 LED CLK LCK

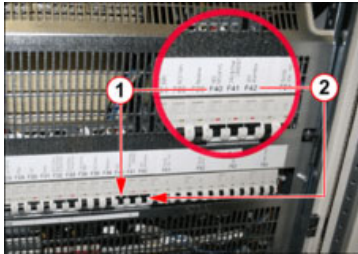


Fig. 538: EPC 36 Volt circuit breakers

- 1 F40 MSUP PS
- 2 F42 Alarm box

EPC cabinet

Switch **OFF** circuit breaker **F40 (1)**.
 The MSUP audible alarm is deactivated.
 The supply LED on the MSUP is **OFF**.

Disconnect the 'D' type electrical connectors.
 the cold head switches **OFF**, if **X1** is disconnected.
 Disconnect the ERDU battery cable from **X3**.  Fig. 537 (Pos. 3)

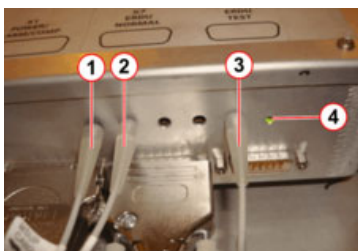


Fig. 539: MSUP 2306 fiber optic cables

- 1 F/O TX1 U5
- 2 F/O RX1 U7
- 3 F/O CLK U9
- 4 LED CLK LCK

Carefully disconnect the (3 off) fiber optic cables.

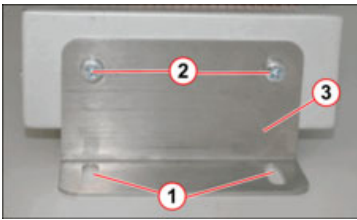


Fig. 540: MSUP 2306 mounting brackets

- 1 Mounting slots
- 2 M6 screws and Nordlock washers (2 off each end)
- 3 Bracket

Remove the (2 off) brackets.

Unscrew the (2 off) M6 screws, spring and flat washers (each end).

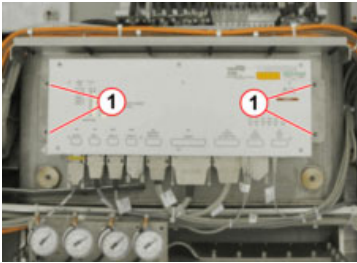


Fig. 541: MSUP 2306 replacing the complete assembly

- 1 M6 nuts and Nordlock washers (2 off) each end

The MSUP is mounted on (4 off) OVC studs.

Remove the (4 off) M6 nuts, spring and flat washers.

Lift the MSUP from the mounting studs.

4.1.6 Installation

4.1.6.1 Installing the MSUP 2306 complete unit

Fit the (2 off) brackets to the new MSUP.

Mount the brackets on the OVC studs.

Secure the MSUP using the (4 off) M6 nuts, spring and flat washers.

i

Do not Short pins on the battery connector.

When connecting the battery cable ensure the pins on the connector are not shorted as this will blow the fuses in the battery pack. The fuses are not replaceable, hence a new battery pack will need to be ordered and fitted.

Connect the 'D' type electrical connectors:
the cold head reactivates.

Carefully reconnect the (3 off) fiber optic cables.  Fig. 539

Reconnect the the ERDU battery cable to **X3**.

Turn ON the circuit breaker **F40** at the EPC.  Fig. 538 (Pos. 1)
On the ETL, ensure the following LEDs are all OFF.

ERDU Test Load	Alarm box
QH1 V LED OFF QH2 V LED OFF SH V LED OFF QH1 I LED OFF QH2 I LED's OFF SH I LED OFF	WARNING OFF Buzzer OFF

4.1.7 Startup

Perform a “Re-start Software” of “Meas&Recon” to download the correct loadware.

i If a download is necessary it will take about 20 min.

4.1.7.1 Testing the MSUP 2306 after replacement

i The Magnet Stop button locks when depressed.
Turn the button clockwise to unlock.

Ensure that the **ERDU test connector** is connected to **X7**.



Fig. 542: Service Magnet Stop button

- 1 Service Magnet Stop button
- 2 Pressure gauge

Service ERDU Stop Button Test

The Service ERDU stop button is located next to the pressure gauge on the magnet.

Check that the test connector is fitted to **X7** (ERDU NORMAL) on the MSUP.

Check that the Service ERDU stop button is not activated (pos.1).

To test the Service ERDU Stop button:

Press the Service ERDU stop button to check that the MSUP audible alarm sounds.

Release the Service ERDU stop button by turning clockwise - The alarm will stop.

The test of the Service ERDU button is completed.

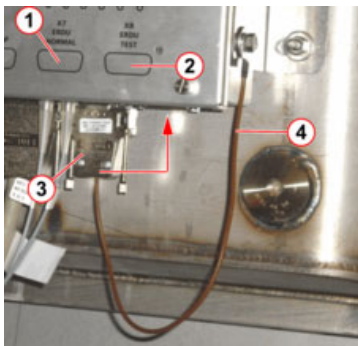


Fig. 543: ERDU test connector

- 1 ERDU NORMAL
- 2 ERDU TEST
- 3 ERDU test connector
- 4 ERDU connector strap

ERDU Button Tests

Use a small flat bladed screwdriver to disconnect the test connector from **X7** (ERDU NORMAL) on the MSUP.

MSUP audible alarm is activated.

Connect it to **X8** (ERDU TEST).

On the alarm box, check the audible alarm sounds.

Press AUDIO ALARM OFF to silence the alarm.

Check that the ERDU TEST LEDs are NOT illuminated on the MSUP.

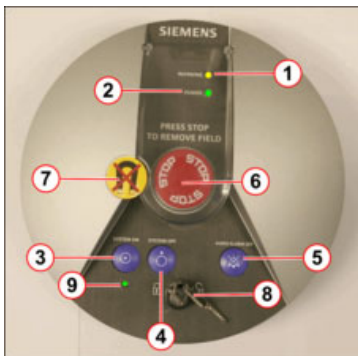


Fig. 544: Alarm box - LEDs and buttons

- 1 LED WARNING = yellow
- 2 LED POWER = green

- 3 Button - SYSTEM ON
- 4 Button - SYSTEM OFF
- 5 Button - AUDIO ALARM OFF
- 6 Button - MAGNET STOP (ERDU)
- 7 Tamper-proof seal
- 8 Power switch locking key
- 9 SYSTEM ON LED = green

At the alarm box:

Remove the tamper-proof seal (if required).

Press the ERDU stop button on the Alarm box. This will not quench the magnet.

Check the audible alarm sounds and the yellow warning LED flashes.

Press AUDIO ALARM OFF to silence the alarm.

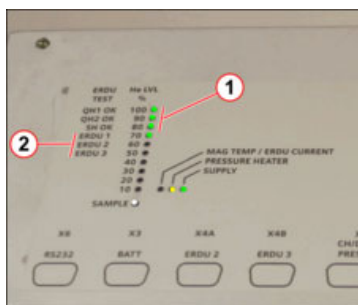


Fig. 545: ERDU test - LED indicators (typical)

- 1 QH1, QH2, SH = green LED (all ON)
- 2 ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

Check **QH1, QH2, SH LEDs** are illuminated.

Check **ERDU 1 LED** is illuminated.

Reset the Magnet Stop button on the alarm box.

Check the **MSUP LEDs** return to normal.

QH1, QH2, SH and **ERDU 1** are **NOT** illuminated.

In the **Magnet Room**:

Press the **Magnet Stop button**.

Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.

Press **AUDIO ALARM OFF** to silence the alarm.

At the **MSUP**:

Check **QH1, QH2, SH LEDs** are illuminated.

Check **ERDU 2 LED** is illuminated.

Reset the **Magnet Stop button** in the Magnet Room.

Check the LEDs return to **normal**.

An optional **3rd Magnet Stop button** may be fitted. Follow the steps below:

Press the **Magnet Stop button**.

Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.

Press **AUDIO ALARM OFF** to silence the alarm.

At the **MSUP**:

Check **QH1, QH2, SH LEDs** are illuminated.

Check **ERDU 3 LED** is illuminated.

Reset the **3rd Magnet Stop button**.

Check the LEDs return to **normal**.

Check **ALL Magnet Stop buttons** are **not activated**.

Disconnect the ERDU test connector **from X8** (ERDU TEST) and reconnect it to **X7** (ERDU NORMAL).

On the alarm box, check the audible alarm sounds.

At the MSUP:

Check the MSUP audible alarm is **deactivated**.

Check the MSUP LEDs return to normal.

At the alarm box:

Press AUDIO ALARM OFF to silence the alarm.

Check the warning LED is not illuminated **'ON'**.

i

If the LED is on, or flashing, the fault must be checked.
Do not disconnect the ERDU battery.

4.1.7.2 Downloading the i-Button MSUP 2306 data to the MRAWP after replacement on Systems with Numaris 4

Ensure that the old data is reading correctly on the new MSUP.

Check the Serial, Revision, and Part numbers of the MSUP.

Update the numbers if necessary.

The **MRC (MRAWP) Host** must be updated with the new Serial, Revision, and Part numbers.

Access the **Service software** at the MRC (MRAWP) Host.

Select menu item **"Magnet & Cooling"**.

Select **"Magnet Setup"**.

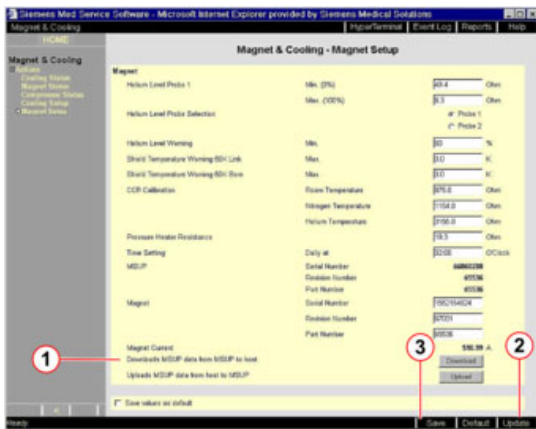


Fig. 546: Updating system Identification numbers

- 1 Download button (MSUP data stored on MRC)
- 2 Update button (refreshes on-screen data)
- 3 Save button (transfer calibration values to MSUP)

Click **“DOWNLOAD”** to transfer calibration values to the **MRC**.

This takes the saved data from the i-Button, and ensures that the MRC data matches the MSUP.

Click **“UPDATE”**.

This refreshes the on-screen values.

4.1.7.3 Downloading the i-Button MSUP 2306 data to the MRAWP after replacement on Systems with Numaris X

Access the Service software at the MRAWP (MRC) Host.

Select menu item “Magnet & Cooling”.

Select “Magnet Setup”.

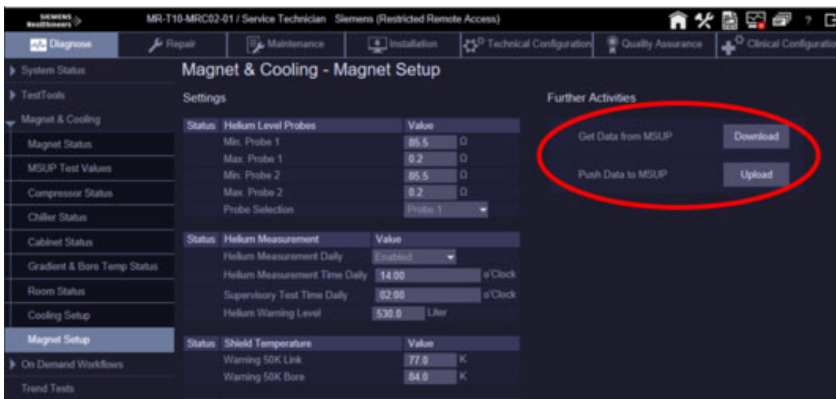


Fig. 547: Uploading data to the MSUP on Systems with Numaris X

Update the serial/revision/part numbers (if required).

Click **“Save”** to transfer calibration values to the MSUP.

4.1.7.4 Restore to normal operation

Disconnect the ETL flying lead from the magnet turret (41-pin connector)

Reconnect the lead from the MSUP (X5) to the magnet turret (41-pin connector).

Perform a **He** test sample.

press the sample button on the MSUP.

wait for **30 secs.**

ensure the LEDs displays the correct He level.

Clear all posted alarms on the MRC.

Press the ACKNOWLEDGE button on the alarm box. Check that all indicators are now normal.

The ERDU test generates some error / warning messages on the MRC. These error messages can be ignored.

Click <OK> to confirm the messages.

4.1.8 Final steps

Disconnect the ETL PSU from the AC supply.

Disconnect the ERDU Test Load from its PSU.

Re-pack all items in the ERDU Test Load Kit to prevent loss.

Replace the appropriate cover panels.

Affix the tamper-proof seal(**Part no.8100831**) to the alarm box.  Fig. 544 (Pos. 7)

4.2 Replacing the Magnet Supervisory Unit Controller Assembly (MSUP 2306)

4.2.1 Requirements

This procedure is applicable to **OR98** and **OR99/OR103** systems.

i

When the system has MSUP faults thus the replacement of the MSUP is required, change the MSUP controller assembly.

4.2.1.1 Safety information

WARNING

Strong magnetic fields.

Failure to observe the following safety warning can result in physical injury and equipment damage.

Only non-magnetic tools and equipment can be taken into the magnet room.

Close the door of the RF room.

4.2.1.2 Estimated completion time and number of persons

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	30 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

Tab. 46: Completion time

4.2.1.3 Tools

Non-magnetic tool kit.

4.2.1.4 Spare parts

MSUP 2306 controller assembly (part no **10128921**).

4.2.2 Preparatory Work Steps

4.2.2.1 General

The MSUP 2306 is a **Field Replaceable Unit**. It contains 2 assemblies:

The MSUP 2306 controller assembly is attached to the underside of the lid and is a replaceable spare part.  Fig. 527 (Pos. 1)

The ERDU assembly is mounted in the base of the MSUP and is not separately replaceable. If faulty, the complete MSUP 2306 unit must be replaced.  Fig. 527 (Pos. 3)

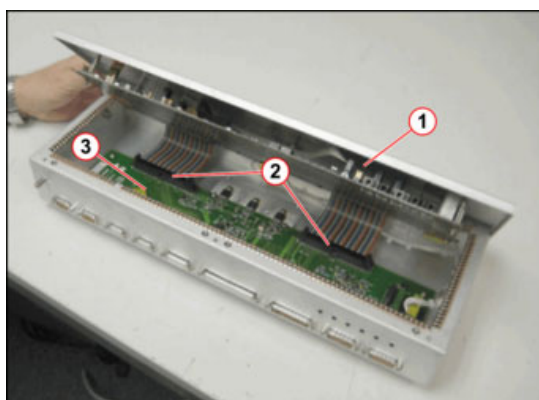


Fig. 548: MSUP 2306 complete unit

- 1 MSUP controller assembly (replaceable part)
- 2 Ribbon cables
- 3 ERDU assembly (not replaceable)

Remove the appropriate cover panels.

4.2.2.2 MSUP 2306 Positioning

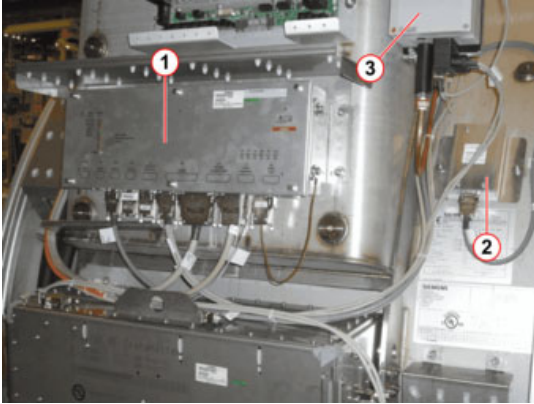


Fig. 549: MSUP 2306 position (static systems)

- 1 MSUP
- 2 ERDU battery
- 3 Pressure gauge

The MSUP 2306 is located behind the magnet service access covers. The unit and battery box are mounted on the magnet OVC. The battery is located to the right of the MSUP 2306.

4.2.3 Removal

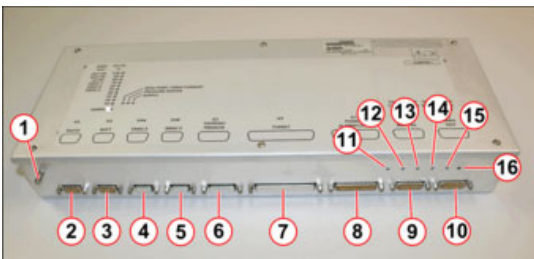


Fig. 550: MSUP 2306 indicators and input/output connections

- 1 Earth stud
- 2 X6 RS232
- 3 X3 Battery
- 4 X4A ERDU 2
- 5 X4B ERDU 3
- 6 X2 CH/DIAG/PRESS/ID
- 7 X5 TURRET
- 8 X1 POWER/ALARM/COMP
- 9 X7 ERDU NORMAL
- 10 X8 ERDU TEST (not connected when system is in use)

- 11 F/O TX1 U5
- 12 F/O RX1 U7
- 13 F/O TX2 U6
- 14 F/O RX2 U8
- 15 F/O CLK U9
- 16 LED CLK LCK

4.2.3.1 Removing the MSUP 2306 controller assembly

i

This procedure can be performed with the magnet at field.

i

When the ERDU test connector is disconnected from BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

Ensure that the **ERDU test connector** is connected to **X7**.



Fig. 551: Service Magnet Stop button

- 1 Service Magnet Stop button
- 2 Pressure gauge

Service ERDU Stop Button Test

The Service ERDU stop button is located next to the pressure gauge on the magnet.

Check that the test connector is fitted to **X7** (ERDU NORMAL) on the MSUP.

Check that the Service ERDU stop button is not activated (pos.1).

To test the Service ERDU Stop button:

Press the Service ERDU stop button to check that the MSUP audible alarm sounds.

Release the Service ERDU stop button by turning clockwise - The alarm will stop.

The test of the Service ERDU button is completed.

⚠ WARNING

Risk of quench.

The magnet can quench if the ERDU test connector is not fitted to X8.

Check ERDU connection on the MSUP.

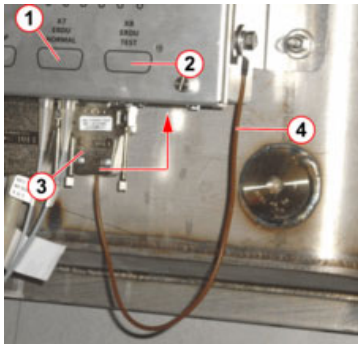


Fig. 552: ERDU test connector

- 1 ERDU NORMAL
- 2 ERDU TEST
- 3 ERDU test connector
- 4 ERDU connector strap

ERDU Button Tests

Use a small flat bladed screwdriver to disconnect the test connector from **X7** (ERDU NORMAL) on the MSUP.

MSUP audible alarm is activated.

Connect it to **X8** (ERDU TEST).

On the alarm box, check the audible alarm sounds.

Press AUDIO ALARM OFF to silence the alarm.

Check that the ERDU TEST LEDs are NOT illuminated on the MSUP.

i

If the MSUP audible warning is not sounding, DO NOT CONTINUE.
Investigate the cause of the problem.

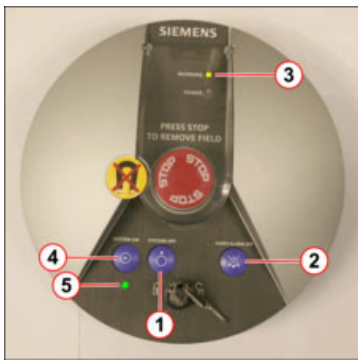


Fig. 553: Alarm box

- 1 Button SYSTEM OFF
- 2 Button AUDIO ALARM OFF
- 3 LED WARNING = yellow
- 4 Button SYSTEM ON
- 5 LED SYSTEM ON = green

Shut down the MRC:

Wait until the MRC has shut down.

Press SYSTEM OFF button on the alarm box (pos 1).

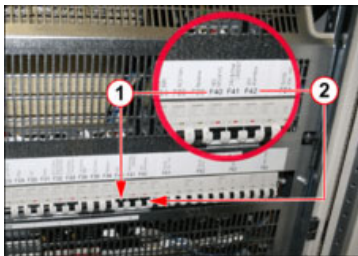


Fig. 554: EPC 36 Volt circuit breakers

- 1 F40 MSUP PS
- 2 F42 Alarm box

EPC cabinet

Switch **OFF** circuit breaker **F40 (1)**.

The MSUP audible alarm is deactivated.

The supply LED on the MSUP is **OFF**.

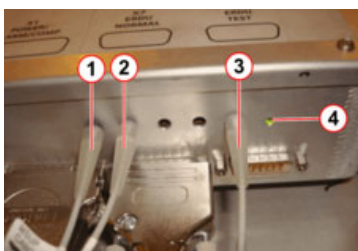


Fig. 555: MSUP 2306 fiber optic cables

- 1 F/O TX1 U5
- 2 F/O RX1 U7

- 3 F/O CLK U9
- 4 LED CLK LCK

Carefully disconnect the (3 off) fiber optic cables.

⚠ CAUTION

Component failure.

ESD precautions must be followed to stop the risk of damaging or reducing the life of the component.

Use ESD ground straps when handling ESD components.

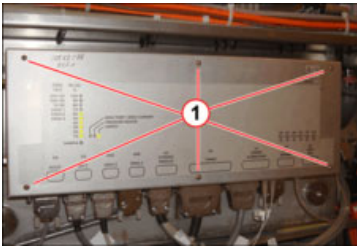


Fig. 556: MSUP 2306 fit/remove the controller assembly

- 1 M3 screws and washers (6 off)

Remove the (6 off) M3 screws securing the MSUP controller assembly.

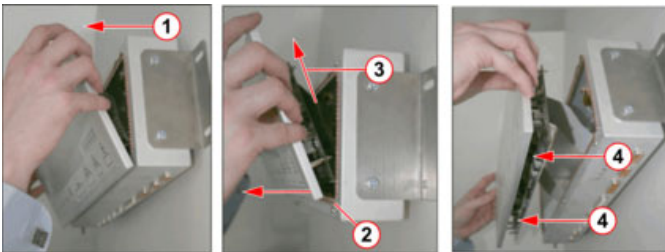


Fig. 557: MSUP 2306 removing the controller assembly

- 1 Lift the lid at 90 degrees
- 2 Lip
- 3 Lift over the lip
- 4 Lift assembly horizontally

Tilt the controller assembly at an 90 degree angle (pos 1).

Lift over the lip (pos 2,3).

Lift the assembly horizontally (pos 4).



Fig. 558: MSUP 2306 controller assembly



Fig. 559: MSUP 2306 ribbon connections

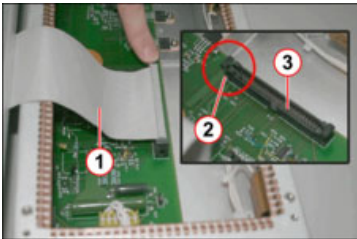


Fig. 560: MSUP 2306 ribbon cables

- 1 Ribbon cable
- 2 Connector clip
- 3 Ribbon cable connector

Release the connector clips  Fig. 560 (Pos. 2) and disconnect the ribbon cables.  Fig. 560 (Pos. 3)

the alarm box audible alarm sounds.

the cold head switches OFF.

The MSUP alarm reactivates.

4.2.4 Installation

4.2.4.1 Fitting the MSUP controller assembly

Fit the ribbon cables.  Fig. 560 (Pos. 1)

Ensure that the connectors are seated correctly.

Lock in place with the connector clips.  Fig. 560 (Pos. 2)

the alarm box audible alarm stops.

the cold head reactivates.

The MSUP alarm stops.



Fig. 561: MSUP 2306 feed through the ribbon cables

Ensure that the ribbon cables are not **trapped** as the controller assembly is re-fitted/
Carefully tuck the ribbon cables under the PCB.

Refit the controller assembly using the (6 off) screws.  Fig. 556 (Pos. 1).
Carefully reconnect the (3 off) fiber optic cables.  Fig. 555

Turn ON the circuit breaker **F40** at the EPC.  Fig. 554 (Pos. 1)

the supply LED on the MSUP is LIT.

the MSUP alarm sounds.

on the alarm box, check the audible alarm sounds.

press AUDIO ALARM OFF to silence the alarm.  Fig. 544 (Pos. 5)

i

The cold head switches off for 1 sec before reactivating.

Test the replacement MSUP. ➤ Testing the MSUP 2306 after replacement

4.2.5 Startup

Perform a **“Re-start Software”** of **“Meas&Recon”** to download the correct loadware.

i

If a download is necessary it will take about 20 min.

4.2.5.1 Testing the MSUP 2306 after replacement

WARNING

Risk of quench.

The magnet can quench if the ERDU test connector is not fitted to X8.

Check ERDU connection on the MSUP.

i

The Magnet Stop button locks when depressed.
Turn the button clockwise to unlock.

Connect the **ERDU test connector** from **X7** to **X8**.

Check that the ERDU TEST LEDs are **NOT** illuminated.

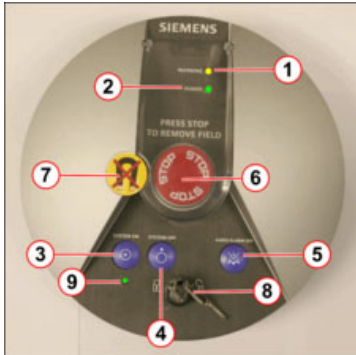


Fig. 562: Alarm box - LEDs and buttons

- 1 LED WARNING = yellow
- 2 LED POWER = green
- 3 Button - SYSTEM ON
- 4 Button - SYSTEM OFF
- 5 Button - AUDIO ALARM OFF
- 6 Button - MAGNET STOP (ERDU)
- 7 Tamper-proof seal
- 8 Power switch locking key
- 9 SYSTEM ON LED = green

At the alarm box:

Remove the tamper-proof seal (if required).

Press the ERDU stop button on the Alarm box. This will not quench the magnet.

Check the audible alarm sounds and the yellow warning LED flashes.

Press AUDIO ALARM OFF to silence the alarm.

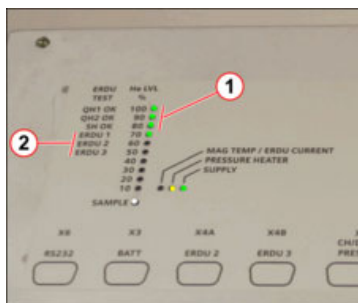


Fig. 563: ERDU test - LED indicators (typical)

- 1 QH1, QH2, SH = green LED (all ON)
- 2 ERDU 1,2,3 = green LEDs (ON only when tested)

At the **MSUP**:

Check **QH1, QH2, SH LEDs** are illuminated.

Check **ERDU 1 LED** is illuminated.

Reset the **Magnet Stop button** on the alarm box.

Check the **MSUP LEDs** return to **normal**.

QH1, QH2, SH and **ERDU 1** are **NOT** illuminated.

In the **Magnet Room**:

Press the **Magnet Stop button**.

Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.

Press **AUDIO ALARM OFF** to silence the alarm.

At the **MSUP**:

Check **QH1, QH2, SH LEDs** are illuminated.

Check **ERDU 2 LED** is illuminated.

Reset the **Magnet Stop button** in the Magnet Room.

Check the LEDs return to **normal**.

An optional **3rd Magnet Stop button** may be fitted. Follow the steps below:

Press the **Magnet Stop button**.

Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.

Press **AUDIO ALARM OFF** to silence the alarm.

At the **MSUP**:

Check **QH1, QH2, SH LEDs** are illuminated.

Check **ERDU 3 LED** is illuminated.

Reset the **3rd Magnet Stop button**.

Check the LEDs return to **normal**.

Check **ALL Magnet Stop buttons** are **not activated**.

Disconnect the ERDU test connector **from X8** (ERDU TEST) and reconnect it to **X7** (ERDU NORMAL).

On the alarm box, check the audible alarm sounds.

At the MSUP:

Check the MSUP audible alarm is deactivated.

Check the MSUP LEDs return to normal.

At the alarm box:

Press **AUDIO ALARM OFF** to silence the alarm.

Check the **warning LED** is not illuminated '**ON**'.

i

If the LED is on, or flashing, the fault must be checked.
Do not disconnect the ERDU battery.

Boot up the MRC:

At the alarm box, press SYSTEM ON.

The green LED will light on the alarm box. (Pos. 5)

Perform a He test sample.

press the sample button on the MSUP.

wait for 30 secs.

ensure the LEDs displays the correct He level.

Clear all posted alarms on the MRC.

4.2.5.2 Checking the magnet calibration data at the Host on Systems with Numaris 4

The MRC (MRAWP) Host must be updated with the new Serial, Revision, and Part numbers, whilst maintaining the magnet calibration figures.

Access the Service software at the MRC (MRAWP) Host.

Select menu item "Magnet & Cooling".

Select "Magnet Setup".

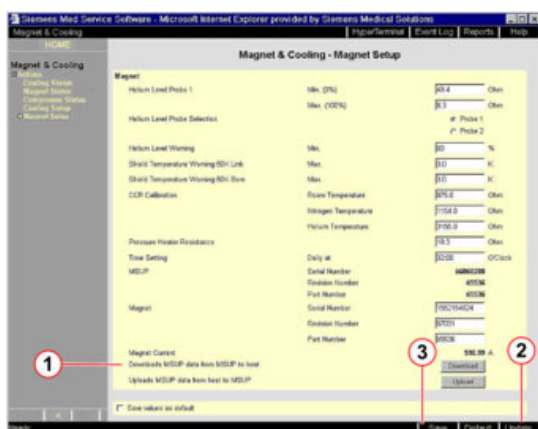


Fig. 564: Updating system Identification numbers

- 1 Download button (MSUP data stored on MRC)
- 2 Update button (refreshes on-screen data)
- 3 Save button (transfer calibration values to MSUP)

Click **"UPDATE"**

This refreshes the on-screen values.

Update any data required, these values should match the host screen shot taken at the start, the only value that will/could be different is the MSUP serial and revision number.

If the data is updated click "SAVE" to transfer the calibration values to the MSUP. Followed by clicking the "Upload" button.

Click the **"DOWNLOAD"** button to ensure everything is now displaying correct information.

4.2.5.3 Checking the magnet calibration data at the Host on Systems with Numaris X

Access the Service software at the MRC (MRAWP) Host.

Select menu item “Magnet & Cooling”.

Select “Magnet Setup”.



(example of an OR105 Avanto Fit (BioFit) system)

Click “**Update**” to update the data on the screen.

4.2.5.4 Checking the magnet calibration data at the MSUP

01 Turn on the laptop and open the **Terminal emulation software**, (eg HyperTerminal).

Set bits per second to 9600.

Set data bits to 8.

Set parity to 1.

Set flow control to none.

Set Emulation to VT100 (File:Properties:Settings)

- 02 **Connect** the laptop to connector **X6** on the MSUP **via RS232 cable**.
- 03 Press **ESC**.
- 04 Type **LO 6932** <enter>.
- 05 Type **R** <enter>.
- 06 Press **ESC** to display the **MSUP Command input screen**.

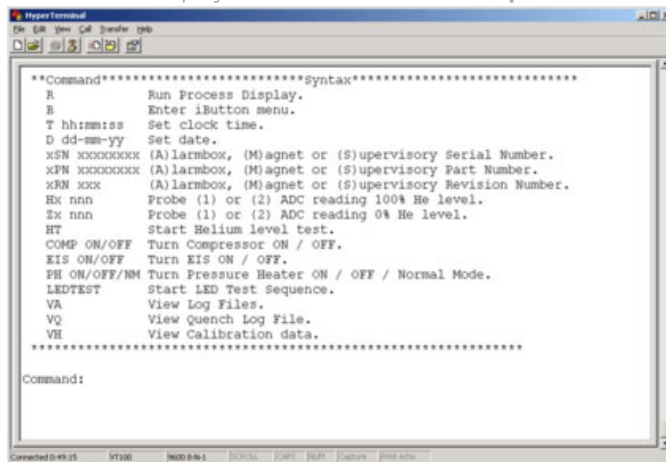


Fig. 566: MSUP Command Input screen (example)

- 07 Type **B** <enter> this displays the **iButton menu**.

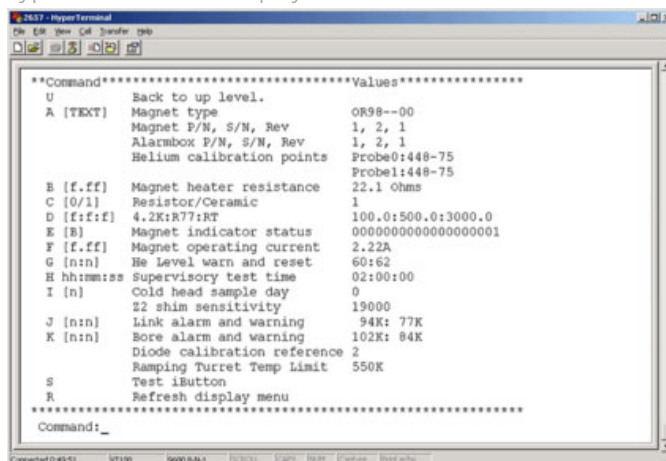


Fig. 567: MSUP 2306 iButton menu (example)

- 08 Using the MSUP screenshot taken at the start of the process update any data on the iButton window to match.
For example, to change the Magnet heater resistance to 22.5 ohms. Type **B 22.5** followed by enter
- 09 Once the data is correct type **U** followed by enter, Then type **R** followed by enter.

4.2.6 Final steps

Replace the appropriate cover panels.

4.3 Replacing the ERDU battery box

4.3.1 Safety information

WARNING

Strong magnetic fields.

Failure to observe the following safety warning can result in physical injury and equipment damage.

Only non-magnetic tools and equipment can be taken into the magnet room.

Close the door of the RF room.

i

The ERDU function can still be maintained if the battery is disconnected, as long as the 36V power input to the MSUP 2306 is maintained.

i

This procedure can be performed with the magnet at field.

4.3.2 Estimated completion time and number of persons

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	15 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

Tab. 47: Completion time

4.3.3 Aids or tools

Tools

Screwdriver (**non-magnetic**).

10mm wrench (**non-magnetic**).

Spare parts

ERDU battery box (part no **10128925**).

4.3.4 Preparatory Work Steps

Remove the appropriate **cover panels**.

4.3.5 Removal

i

Disconnecting/connecting the battery cable can cause the alarm box to alarm.
Press AUDIO ALARM OFF to cancel the alarm.

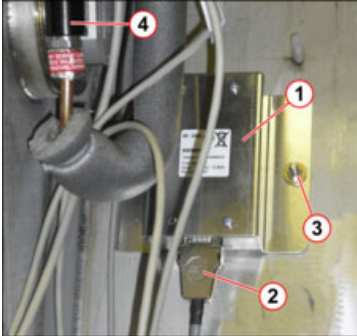


Fig. 568: ERDU battery box location

- 1 ERDU battery box
- 2 ERDU battery cable
- 3 Nut and washer (2 off)
- 4 Pressure gauge

Disconnect the **battery cable**.

This may cause the **alarm box** to sound.

Press **AUDIO ALARM OFF** to silence the alarm.

Remove the nuts and washers (2 off).

Remove the old battery box.

4.3.6 Installation

Fit the new **battery box** with the nuts and washers.

i

Do not Short pins on the battery connector.

When connecting the battery cable ensure the pins on the connector are not shorted as this will blow the fuses in the battery pack. The fuses are not replaceable, hence a new battery pack will need to be ordered and fitted.

Connect the **battery cable**.

4.3.7 Startup

Ensure that there are **no alarms**.

Fault find as required.

4.3.8 Final Steps

Replace the appropriate **cover panels**.

